

COMPUTER VISION (A.A. 2025–2026)

Sapienza Università di Roma

Emerging Interpretability in LeJEPA vs. Supervised Learning

Investigating Zero-Shot Latent Semantics through Spatial PCA and XAI Saliency

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Presentation Outline

Roadmap of the comparative study.

01

Motivation & Context

- **Problem Statement:** Latent Semantics vs. Shortcut Learning
- **State of the Art:** Self-Supervised JEPAs & Model Explainability

02

Proposed Methodology

- **Spatial PCA:** Unsupervised Latent Object Projections
- **Semantic Alignment Score (SAS):** Quantitative XAI-PCA Correlation

03

Experimental Protocol

- **Standardized CIFAR-10** Benchmarking & Intersecting Subsets
- **Trajectory Probing &** Relative Accuracy Alignment

04

Evaluation & Discussion

- **Qualitative Saliency** vs. Latent Semantic Overlays
- **Layer-by-Layer** Alignment Curves & Metric Agreement
- **Key Takeaways &** Future Directions

Problem Statement

Shortcut Learning vs. Emergent Semantics

Supervised Cross-Entropy (Shortcut Bias)

Objective: Optimizes exclusively for categorical class boundaries.

Representational Flaw: Incentivizes sparse, discriminative shortcuts (textures, backgrounds, isolated edges) rather than whole-object semantics.

LeJEPa & SIGReg (Emergent Semantics)

Objective: Enforces multi-view latent invariance without labels or negative pairs.

Hypothesis: Non-collapse constraints force the latent space to organize visual scenes into cohesive, zero-shot object segments.

THE CORE QUESTION

"Do self-supervised latent features naturally align with operational decision-making, or do they exhibit the same localized biases as supervised models?"

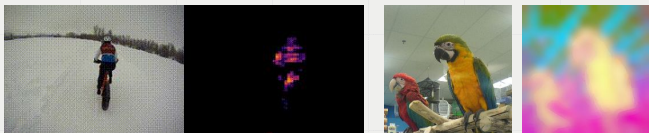
State of the Art

LeJEPa and XAI

Self-Supervised Learning — LeJEPa & SIGReg (Balestriero & LeCun, 2025)

Provable SSL: Learns representations via multi-view invariance and Sketched Isotropic Gaussian Regularization without contrastive negative pairs or heuristic stop-gradients.

Emergent Semantics: Latent space naturally structures coordinates into zero-shot semantic segments without dense pixel labels.



Emergent zero-shot semantic segmentation visible via latent PCA projections without pixel-level supervision.

Explainable AI Baselines (XAI)

Grad-CAM (Selvaraju et al., 2017): Calculates target-class gradients pooled across convolutional channels to localize class-discriminative evidence in CNNs.

GMAR (Jo et al., 2025): Gradient-Driven Multi-Head Attention Rollout across transformer encoder blocks, weighting attention heads by gradient importance.

CNN Feature Attribution
Grad-CAM Gradient Pooling

ViT Attention Rollout
GMAR Token Attribution

Architectural comparison schematic: CNN feature attribution vs. ViT gradient-weighted attention rollout.

Model Architecture & Training Paradigms

The 2x2 Comparative Matrix

DUAL VISION ARCHITECTURES

ResNet18 (CNN)

4 convolutional residual stages (64, 128, 256, 512 channels) with a 3×3 stride-1 stem tailored for 32×32 spatial feature preservation; enforces local inductive biases.

Vision Transformer (ViT)

6 encoder blocks, embedding dimension 512, 8 attention heads, and 4×4 patch projection (64 spatial tokens + 1 CLS token); processes global relational context.

RESNET18

ResNet18
+ SUPERVISED

ResNet18
+ LEJEPA (SSL)

VIT (TRANSFORMER)

ViT
+ SUPERVISED

ViT
+ LEJEPA (SSL)

SUPERVISED

LEJEPA (SSL)

DUAL LEARNING PARADIGMS

Supervised Learning (Baseline)

End-to-end categorical cross-entropy classification driving task-specific discriminative representations.

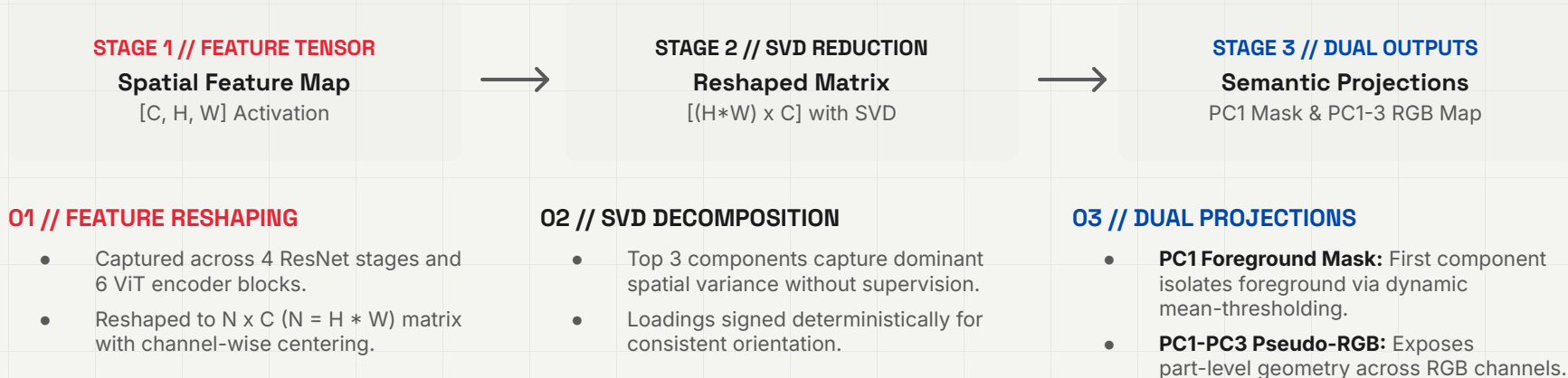
LeJEPA (Self-Supervised)

Latent predictive learning across multi-view crops (2 global, 4 local) optimized via representation invariance and Sketched Isotropic Gaussian Regularization (SIGReg) to prevent collapse.

Evaluating identical architectures across supervised and self-supervised regimes isolates the direct impact of the learning objective on internal latent structures.

Spatial PCA Latent Decomposition

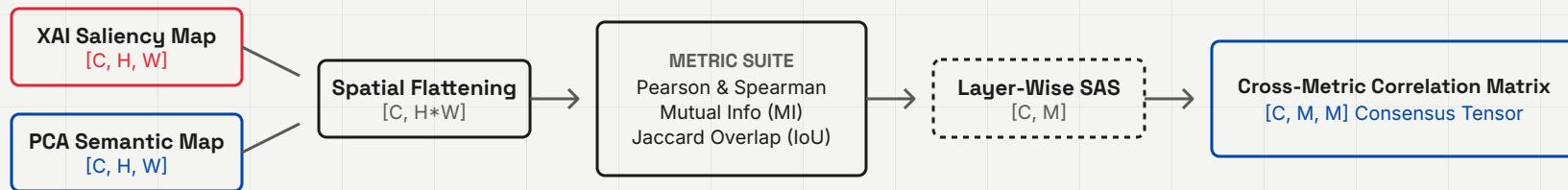
Unsupervised Extraction of Emergent Semantics Across Depth



Spatial PCA turns internal feature activation into interpretable visual masks without requiring pixel annotations or fine-tuning.

Decision Saliency & SAS Framework

Quantifying the Alignment Between Decision Saliency and Latent Semantics



01 // ARCHITECTURE-TAILORED XAI

- **Grad-CAM for CNNs:** Pools target-class gradients across convolutional feature maps to capture class-discriminative localization.
- **GMAR for ViTs:** Progressively rolls out gradient-weighted multi-head attention across transformer blocks, using L1 token gradient norms to retain decision flow.

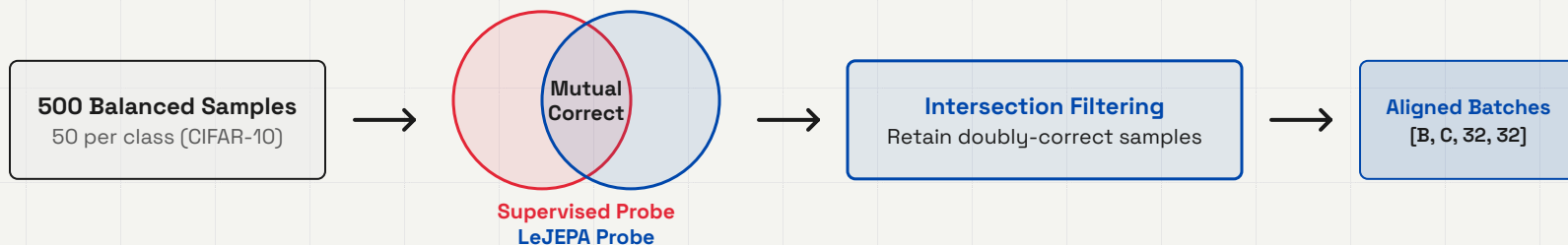
02 // THE SEMANTIC ALIGNMENT SCORE (SAS)

- **Intra-Model Alignment:** Measures operational decision alignment with latent geometry: $SAS = M(XAI_layer, PCA_layer)$.
- **Multi-Metric Suite:** Combines linear Pearson, monotonic Spearman rank, non-linear MI, and localized Jaccard overlap (IoU).
- **Metric Agreement Matrix:** Evaluates Pearson correlation across metrics over the batch to produce a [C, M, M] consensus tensor.

SAS bridges operational decision attribution with latent geometric representation, evaluated layer-by-layer across independent statistical metrics.

Dataset & Evaluation Setting

Standardized CIFAR-10 Benchmarking & Controlled Intersection Filtering



BENCHMARK CONFIGURATION & SUBSET

- **CIFAR-10 Scale:** 10 semantic classes evaluated at native 32×32 image resolution.
- **500-Sample Evaluation Subset:** Deterministically selects 50 balanced test samples per class using fixed pseudo-random seeds.

THE INTERSECTION FILTERING LOGIC

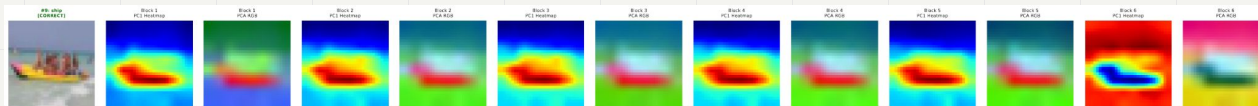
- **The Gradient Validity Challenge:** Gradient attribution relies on ground-truth logit scores; misclassified images generate uninformative, noisy heatmaps.
- **Deterministic Intersection:** Only images correctly predicted by both models (Supervised & LeJEPA Linear Probe) are retained.

Qualitative Analysis

Saliency & Latent Semantics Across Depth

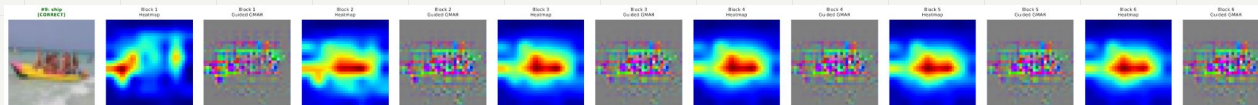
ViT Spatial PCA

PC1 Mask & Pseudo-RGB (Blocks 1–6)



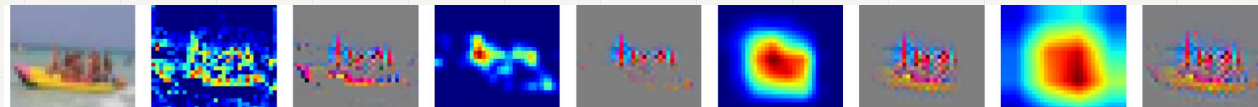
ViT GMAR

Guided Attention Rollout (Blocks 1–6)



CNN Grad-CAM

& Guided CAM (Stages 1–4)



INDUCTIVE BIAS VS. GLOBAL ATTENTION

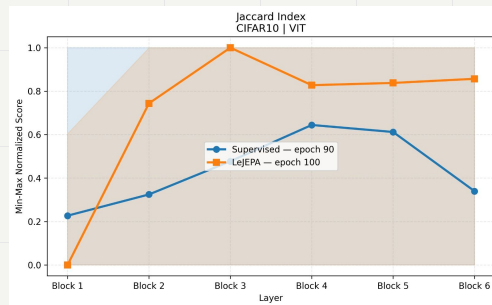
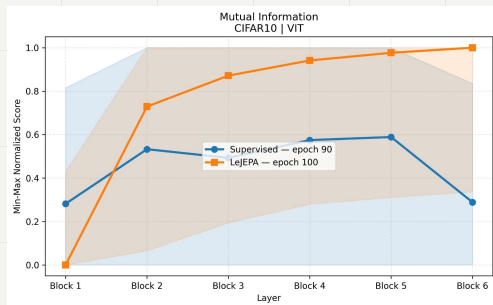
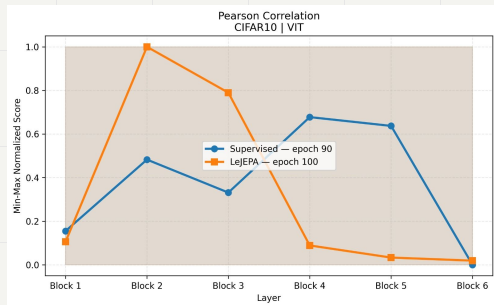
CNN feature attribution gradually builds receptive field coverage stage-by-stage, whereas ViT self-attention captures whole-object context immediately in shallow blocks.

LATENT SALIENCY CONGRUENCE

In ViT intermediate blocks (Blocks 2–4), GMAR attention hotspots visually mirror the exact object geometry delineated by the unsupervised PC1 semantic mask.

Quantitative Results: ViT

Accelerated Early Semantic Emergence & Non-Linear Spatial Shift



EARLY EMERGENCE (BLOCKS 2-3)

Semantic alignment emerges substantially earlier in the transformer hierarchy under LeJEPA, outscoring supervised learning across all evaluated metrics simultaneously (linear correlation, rank correlation, Jaccard overlap, and reconstruction error).

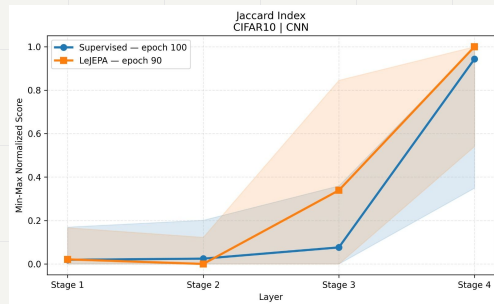
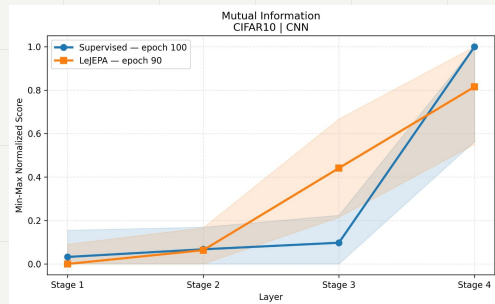
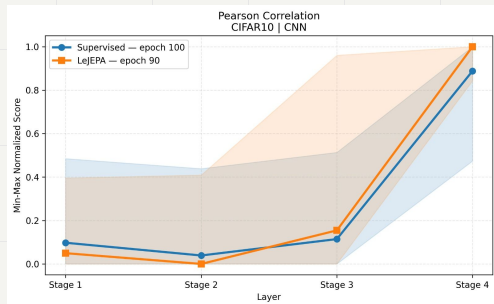
DEEP NON-LINEAR REORGANIZATION (BLOCKS 4-6)

While spatial linear correlation drops in late blocks, Mutual Information and Jaccard overlap remain dominant, showing that LeJEPA preserves semantic content while progressively reorganizing spatial geometry for task processing.

Self-supervised predictive learning accelerates semantic formation to shallow transformer blocks and retains high information fidelity in deeper layers.

Quantitative Results: CNN

Hierarchical Late-Stage Emergence Driven by Spatial Receptive Fields



RECEPTIVE FIELD HIERARCHY (STAGES 1-2)

Early convolutional layers are constrained by small local receptive fields, focusing on low-level textures with near-zero alignment across both paradigms.

LATE BREAKOUT & STRONG CONVERGENCE (STAGES 3-4)

LeJEPA initiates semantic alignment earlier at Stage 3 and achieves superior spatial overlap (Jaccard), lowest reconstruction error (MSE), and strong linear alignment in the final stage.

Convolutional architectures enforce a strict spatial hierarchy where semantic alignment is delayed until receptive fields span the entire object in deep stages.

Conclusions

Emergence Dynamics & Representation Insights

01 // VALIDATION OF EMERGENT SEMANTICS

- **Hypothesis Confirmed:** LeJEPa's predictive invariance and SIGReg regularization foster genuine zero-shot semantic organization, avoiding the isolated shortcut bias of cross-entropy supervision.
- **Decision Alignment:** Latent semantic structures reflect operational saliency attributions significantly more coherently than supervised baselines.

02 // ARCHITECTURAL BIAS & TIMING

- **ViT Acceleration:** Global self-attention enables rapid, early-stage semantic emergence (peaking in Blocks 2–3 across all evaluation metrics).
- **CNN Spatial Hierarchy:** Local convolutional inductive biases strictly delay semantic alignment until late layers (Stages 3–4), once receptive fields span holistic objects.

03 // INFORMATION BEYOND LINEARITY

- **Non-Linear Reorganization:** While spatial linear correlation diminishes in deep transformer blocks, Mutual Information remains dominant.
- **Metric Sensitivity:** Multi-metric evaluation is vital; linear metrics alone overlook deep transformer feature reorganization that preserves semantic content.

FUTURE DIRECTIONS // Scaling validation from CIFAR-10 (32×32) to high-resolution ImageNet benchmarks (224×224) and evaluating against dense ground-truth segmentation masks.

References

Foundational Literature & Experimental Frameworks

Balestriero, R., & LeCun, Y. (2025). *LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics*. arXiv preprint.

Jo, S., Jang, G., & Park, H. (2025). *GMAR: Gradient-Driven Multi-Head Attention Rollout for Vision Transformer Interpretability*. Proceedings of Computer Vision Conferences.

Selvaraju, R. R., Cogswell, M., Das, A., Vedaldi, A., Parikh, D., & Batra, D. (2017). *Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization*. IEEE International Conference on Computer Vision (ICCV).

Code, cached activations, and evaluation pipelines available on GitHub.

Thank you — Open for Questions

